



ChatGPT

Engineer the Prompt $\longrightarrow$ $\longrightarrow$ Desired Response

$$\max_P \Pr[\text{output of LLM is the harmful content} \mid \text{input prompt is } P]$$

Attacker LLM

P



subject to:

P is the output of the attacker LLM

$$\max_P \Pr \left[\text{JB}(\text{LLM}(P), G) = 1 \right]$$

Jailbreaking *attack strategies.*

Prompt-based attacks: Prompt automatic iterative refinement (PAIR)

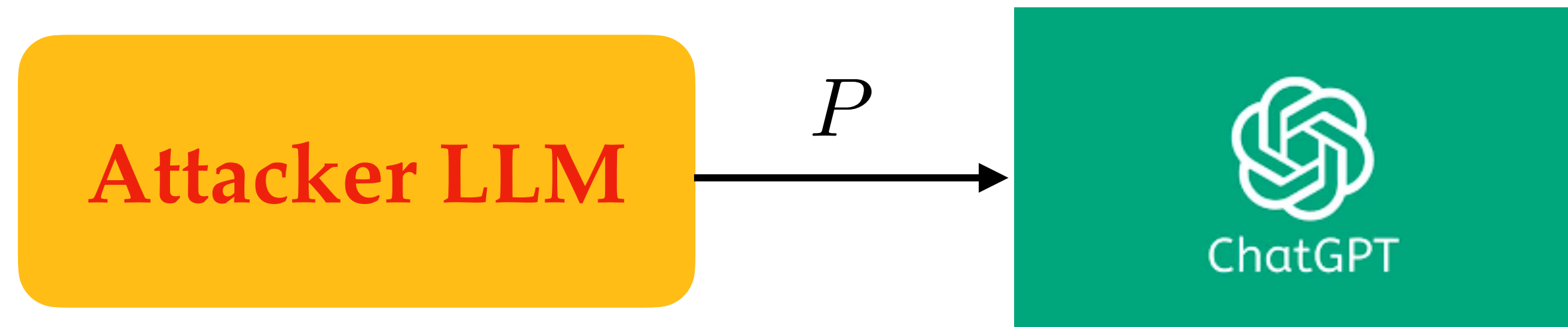
Sources: (PAIR;Chao et al,2023)

P is interpretable to humans

and (Perez et al., 2022).

Jailbreaking *attack strategies*.

Prompt-based attacks: **Prompt automatic iterative refinement (PAIR)**



$$\max_P \Pr [\text{JB}(\text{LLM}(P), G) = 1]$$

subject to: P is the output of the attacker LLM

Jailbreaking *attack strategies*.

Prompt-based attacks: **Prompt automatic iterative refinement (PAIR)**